Effect of Hydrogen Dilution on the Limiting Mechanism for Cerium Hydriding

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Abstract

The question examined here is whether hydride formation on a cerium particle exposed to an H_2/Ar atmosphere remains governed by the same limiting mechanism as the hydrogen concentration is reduced. This question is motivated by a practical experimental need: hydrogen diluted in argon would be substantially safer to use than pure hydrogen, but the resulting data are only useful if dilution does not fundamentally change the controlling physics. A quasi-steady spherical model is developed in which hydrogen uptake is limited by three serial processes: gas-phase transport through H_2/Ar, transport through a hydride shell, and interfacial reaction at the Ce/CeH_2 front. The model is used to map the dominant bottleneck as a function of temperature, total pressure, hydrogen mole fraction, and particle size. Over the range examined here, shell diffusion is often the dominant transport resistance and kinetics becomes increasingly important at higher temperature and pressure. Gas-phase diffusion seems to not be a factor except for extremely low concentrations. These results suggest that hydrogen dilution does not merely reduce the overall hydrogen availability; but can also shift the limiting behavior. Consequently, the use of dilute H_2/Ar mixtures in future experiments is most defensible when the operating conditions remain within the same limiting regime as the pure-hydrogen case of interest.

1 Introduction

The hydriding reaction of cerium is governed by coupled transport and reaction processes. A hydrogen molecule must first reach the particle through the surrounding gas, then diffuse through any hydride layer that has already formed, and finally react at the Ce/CeH_2 interface. The relative rates of these steps determines the limiting behavior and define the extent to which laboratory conditions can be varied without changing the underlying physics.

The original proposal for this project focused on a characteristic hydriding time for a hot liquid cerium droplet and asked whether the process was controlled primarily by gas-phase transport or by interfacial reaction. As work on the project progressed, two major difficulties motivated a change of focus. First, a defensible "characteristic hydriding time" prediction would require a fully coupled model for heat transfer back to the surface. Second, once a hydride layer is included, diffusion through that layer becomes a separate candidate bottleneck and cannot be folded into a generic “transport-limited” category. These are combined with the inability to find an accurate kinetic model for the hydriding reaction of molten cerium. The present report therefore addresses a related question: ***under fixed thermodynamic conditions, is cerium hydriding controlled primarily by gas-phase diffusion, shell diffusion, or interfacial kinetics, and how does that classification change as hydrogen is diluted with argon?***

This refinement is mostly a matter of modeling convenience. In planned experimental work, using hydrogen diluted in argon would offer a substantial safety advantage relative to pure hydrogen. However, diluting the mixture risks changing the limiting mechanism as compared to the conditions of real interest. If dilution leaves the controlling regime unchanged, then reduced-H_2 experiments may still provide mechanistically relevant data. If dilution shifts the system from shell-diffusion control to gas-diffusion control, or from transport control to kinetic control, then the results would have to be interpreted much more cautiously.

The objective of this project is therefore to quantify how the dominant bottleneck depends on hydrogen mole fraction, total pressure, temperature, and particle size. The model is intentionally quasi-steady and isothermal. It is not intended to predict a full transient hydriding history; rather, it is meant to identify the conditions under which hydrogen dilution is likely to preserve or alter the controlling mechanism. Within that framework, the shell-diffusion expression is taken from the reduced hydride-layer model of Regele et al.[1] and the interfacial kinetics term is taken from Sarussi et al.[2] despite it's lack of applicability to molten cerium.

2 Model Formulation

2.1 Physical picture

The system is represented as a single spherical cerium particle of core radius a surrounded by a layer of cerium hydride of thickness δ, this is surronded in turn by a H_2/Ar gas mixture. The radius of the solid phase is defined as b so that $b = a + \delta$. At a given operating point, hydrogen uptake is modeled as a sequence of three serial steps:

1. diffusion through the surrounding gas from the far field to the outer particle surface,
2. diffusion through the hydride shell from the outer surface to the metal/hydride interface, and
3. reaction at the Ce/CeH_2 interface.

The coupled uptake rate is determined by enforcing equality of the hydrogen flux through all three steps.

2.2 Gas-phase transport

External gas transport is modeled with a Stefan-corrected spherical expression for inward diffusion of H_2 through stagnant Ar:

$$\dot{n}_g = 4\pi c D_{H_2-Ar} \frac{\ln[(1-y_s)\,/\,(1-y_\infty)]}{(1/b - 1/R_\infty)}, \tag{1}$$

where y_∞ is the far-field hydrogen mole fraction, y_s is the hydrogen mole fraction at the outer particle surface, D_{H_2-Ar} is the binary diffusion coefficient obtained from Cantera, and $c = P/(RT)$ is the total molar concentration. Near pure hydrogen the stagnant-carrier form becomes singular, so the gas-side resistance is neglected in that limiting case.

2.3 Shell transport and interfacial kinetics

Hydrogen transport through the hydride shell is represented by a reduced spherical diffusion law,

$$\dot{n}_{sh} = 4\pi D_{sh} \frac{ab}{b-a}\, c_{\mathrm{ref}}\,(y_s - y_i)\,, \tag{2}$$

following the spherical hydride-layer resistance form used by Regele et al.[1] In that model, the hydride shell is treated as a diffusive barrier between the outer particle surface and the reaction front, so the shell resistance scales with the spherical geometric factor $ab/(b-a)$ rather than with a simple planar thickness alone. The quantity y_i is an effective shell-side hydrogen driving variable at the inner interface and D_{sh} is an effective shell diffusivity. This shell variable is not a literal gas-phase mole fraction inside the solid. Instead, it should be interpreted as a normalized hydrogen state variable introduced to represent the driving force for diffusion through the hydride layer in the reduced model of Regele et al.[1] Written this way, the shell-diffusion term can be coupled transparently to the gas-transport resistance and the interfacial kinetics term.

The interfacial hydrogen consumption rate is written in terms of a front speed law,

$$\dot{n}_k = 4\pi a^2 \left(\frac{\rho_{\mathrm{Ce}}}{M_{\mathrm{Ce}}}\right) U(p_{H_2}, T), \tag{3}$$

where U is obtained from the pressure- and temperature-dependent fit reported by Sarussi et al.[2] Although that law was developed from solid-cerium data, it is used here as an effective placeholder kinetics model for the unresolved interfacial reaction.

2.4 Bottleneck classification

To determine the dominant limiting mechanism, the model evaluates three one-process capacities:

1. a gas-only capacity in which the particle surface is treated as a perfect sink,
2. a shell-only capacity in which the shell sees the full upstream hydrogen level and a perfect sink at the inner interface, and
3. a kinetics-only capacity in which the interface is exposed to the full upstream hydrogen level.

The reciprocals of these capacities are interpreted as resistance-like quantities. The largest resistance share identifies the dominant bottleneck. This construction makes it possible to distinguish explicitly among gas diffusion, shell diffusion, and kinetics to ensure that any reduced hydrogen experiments remain in the same limiting regime

3 Baseline Conditions

Table 1 lists the baseline values used to generate the preliminary maps and profile comparisons shown below.

Table 1: Baseline parameters used in the quasi-steady calculations.

Quantity	Value
Particle diameter	50 µm
Hydride shell thickness	100 nm
Shell diffusivity (solid)	$4 \times 10^{-8}\,\mathrm{m^2\,s^{-1}}$
Shell melt threshold	1690 K
Reference total pressure	1 atm
Representative temperature	1450 K
Representative far-field y_{H_2} values	0.002, 0.02, 0.2

4 Results and Discussion

Figure 1 shows the dominant bottleneck as a function of temperature and far-field hydrogen mole fraction for three particle diameters at 1 atm. Two features are immediately evident. First, the dominant competition over most of the map is between shell diffusion and kinetics. Second, particle size has a relatively modest effect on the overall topology of the map for the baseline shell parameters used here. Gas-phase diffusion appears only in a narrow low-H_2 region, which indicates that under many conditions the external gas supply is faster than transport through the shell. Note that above 1690 K CeH_2 melts and consistent with Regele et al. [1] we increase the diffusion coefficient by an order of magnitude, this results in the abrupt change at that temperature.

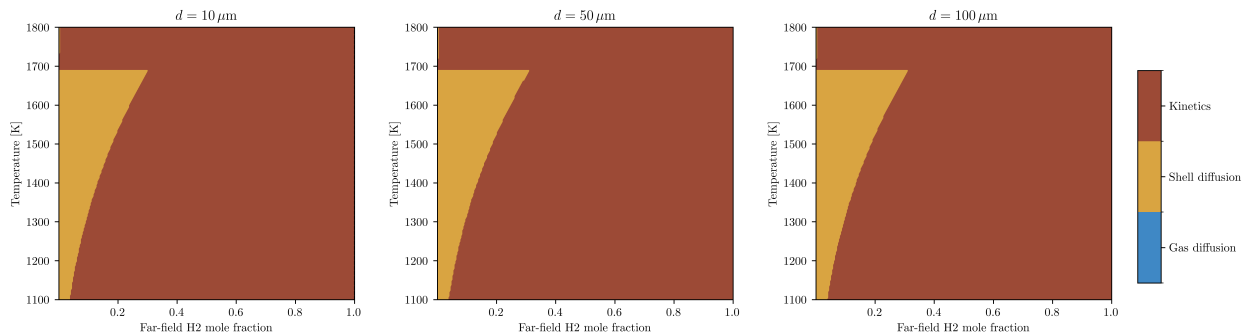

Figure 1: Dominant limiting mechanism as a function of temperature and far-field hydrogen mole fraction for three particle diameters at 1 atm. The color bar identifies whether the controlling resistance is gas diffusion, shell diffusion, or interfacial kinetics.

Figure 2 addresses the question most directly relevant to the use of hydrogen diluted in argon. When the total pressure is reduced, the gas-diffusion contribution grows and the transport-controlled region expands. When the total pressure is increased, the gas-side resistance becomes less important and the map shifts toward kinetic control. Therefore hydrogen dilution cannot be interpreted solely as a reduction in hydrogen availability. Its effect depends on how dilution is introduced and on the accompanying total pressure. A dilute mixture may still be mechanistically useful, but only if the chosen operating point remains in the same regime as the higher-hydrogen case of interest.

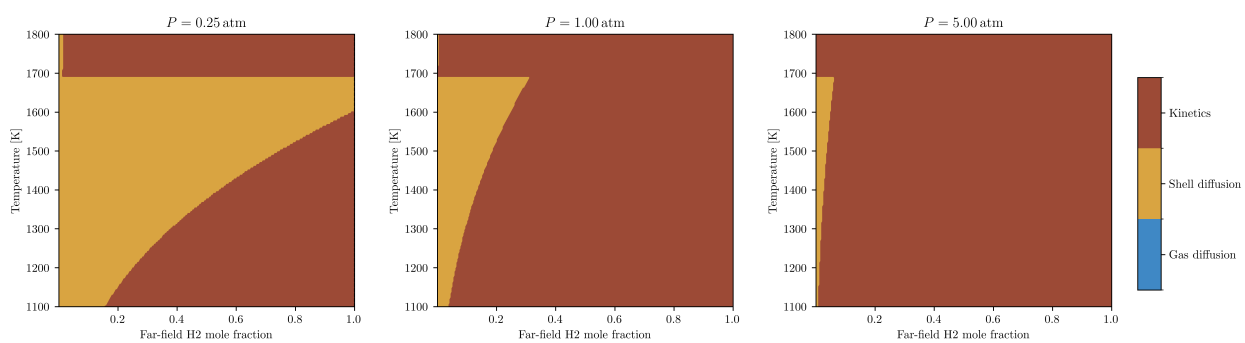

Figure 2: Effect of total pressure on the dominant limiting mechanism for a 50 µm particle. Lower total pressure broadens the region in which transport, particularly shell diffusion and gas diffusion, is important, whereas higher total pressure favors kinetic control in the present model.

The quasi-steady radial profiles are shown in Fig. 3 for three far-field hydrogen mole fractions

at 1450 K and 1 atm. The figure is divided into separate metal, shell, and gas regions so that each spatial scale can be resolved clearly. Inside the metal, the hydrogen level is taken to be zero in the present reduced description. Inside the shell, the dashed curves represent the effective shell-side hydrogen variable used in the diffusion model. Outside the particle, the solid curves show the gas-phase hydrogen mole fraction. For the conditions shown here, the gas-phase profile is nearly flat while the shell exhibits a much larger gradient, reinforcing the conclusion that the shell resistance is often the dominant transport contribution. Increasing the far-field hydrogen mole fraction raises both the shell-side and gas-side levels, but it does not eliminate the shell drop.

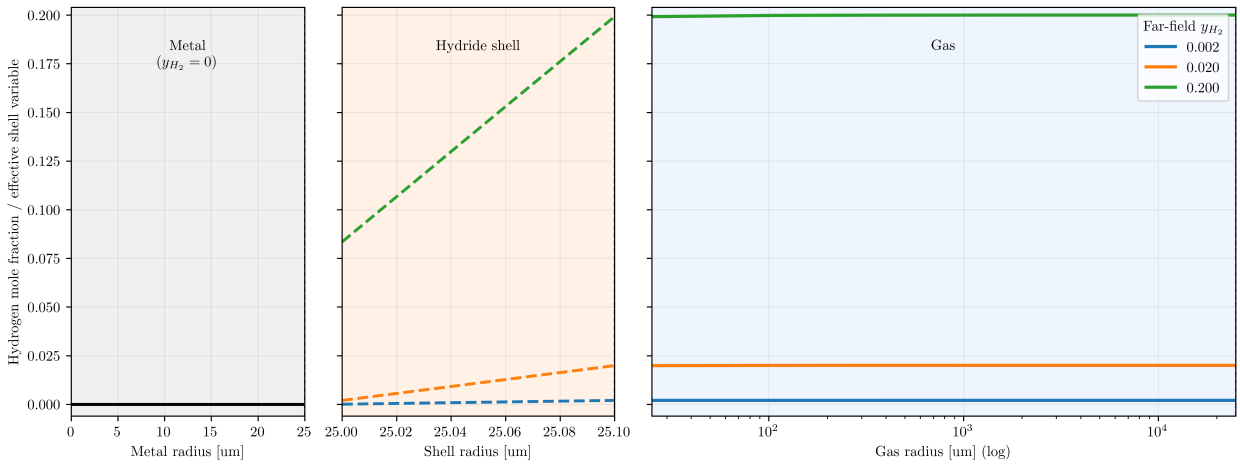

Figure 3: Quasi-steady hydrogen profiles in the metal, shell, and gas regions for three far-field hydrogen mole fractions at 1450 K and 1 atm. Solid lines denote the gas-phase hydrogen mole fraction and dashed lines denote the effective shell-side hydrogen variable for the same condition.

Taken together, these results support a practical interpretation for future experimental design. If dilute H_2/Ar conditions are selected so that the operating point remains within the same limiting region as the higher-hydrogen condition of interest, then dilution may provide a safer route to performing still mechanistically relevant experiments. If, however, dilution shifts the system into a different controlling regime, then the resulting data may reflect a different balance of physics and should not be treated as a simple surrogate for the less diluted case.

5 Limitations

Several limitations should be emphasized. The model is quasi-steady and isothermal, so it does not resolve thermal feedback from the reaction. The hydride shell is represented by an effective diffusion law rather than a detailed defect-transport model. The shell thickness is prescribed in the baseline calculations rather than determined from a fully coupled moving-boundary problem. Most importantly, the interfacial kinetics law is likely invalid since it is based on solid-cerium data.[2] The present results should therefore be interpreted primarily as a mechanistic regime study rather than as a predictive model of absolute hydriding rates. If a more accurate liquid hydriding kinetic mechanism can be found, replacing the existing model will be relatively trivial and the resulting predictions will be much more accurate.

6 Conclusions

The analysis presented here addresses whether hydrogen dilution in an H_2/Ar mixture changes the limiting mechanism for cerium hydriding. Within the quasi-steady framework used here, the answer is that dilution can change the balance among gas diffusion, shell diffusion, and kinetics, especially when it is accompanied by changes in total pressure. Shell diffusion is frequently the dominant transport resistance for the baseline conditions examined, but kinetics can become controlling at sufficiently high temperature and pressure, while gas-phase diffusion becomes more relevant as total pressure decreases. The practical implication is that dilute hydrogen experiments may still be useful for future research, but only if the experimental conditions are chosen so that the same limiting regime is preserved and the results are interpreted carefully.

References

[1] Regele, J. D., Ouellet, F., and Harrison, A. K., "A hydride model for solid and liquid cerium ejecta particles," Tech. Rep. LA-UR-23-30817, Los Alamos National Laboratory, Sept. 2023, Approved for public release; distribution is unlimited.

[2] Sarussi, D., Jacob, I., Bloch, J., Shamir, N., and Mintz, M. H., "The kinetics and mechanism of cerium hydride formation," *Journal of Alloys and Compounds*, Vol. 191, 1993, pp. 91–99.